Winning baseball games by solving statistics puzzles

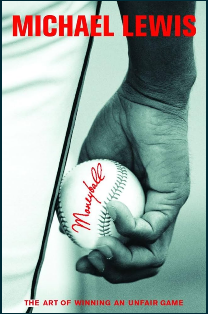
Scott Powers

TAMIDS Seminar Series
February 16, 2026

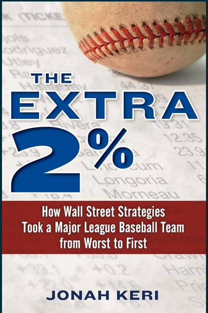
RICE UNIVERSITY
Sport Analytics

What is Sport Analytics?

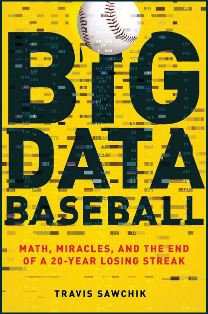
Books about Baseball Teams Using Analytics to Win



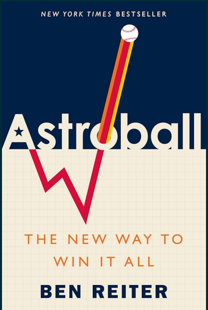
2002
Oakland
Athletics



2008
Tampa Bay
Rays



2013
Pittsburgh
Pirates

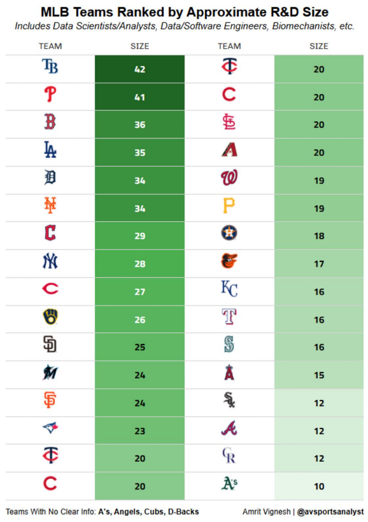


2017
Houston
Astros

MLB R&D Staff Sizes (according to Reddit)

Team	Analysts	Team	Analysts
American League		**National League**	
Yankees	20	Dodgers	20
Astros	15	Braves	15
Rays	15	Brewers	11
Angels	10	Reds	11
Tigers	9	Phillies	10
Rangers	8	Nationals	8
Mariners	7	Pirates	8
Royals	7	Padres	7
Twins	7	Cardinals	6
Blue Jays	6	Cubs	6
Red Sox	6	Giants	6
Indians	5	Marlins	6
Orioles	5	Dbacks	5
Athletics	3	Rockies	4
White Sox	2	Mets	3

2018

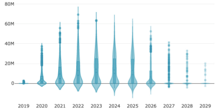


2025

What are they all doing?

Player Evaluation

Executives use modeling results to decide which players to target in trades and in free agency.

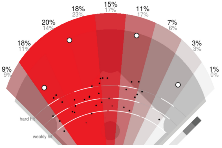


Player Development

Players use data-driven insights to tweak their game and achieve greater levels of performance.

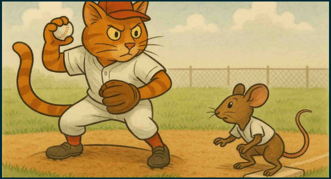
In-Game Strategy

Coaches use analytics tools to position fielders and decide when to make substitutions.



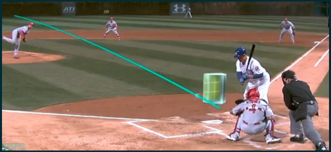
Three Puzzles

I: How far should you stray
from first base?



II: Should you just swing
as hard as you can?

III: What does a pitch
tell you about a pitcher?



How far should you stray from first base?

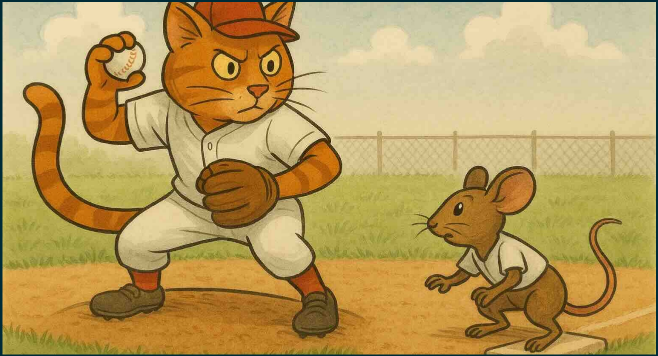

Sivaramakrishnan Ramani, Rice University


Jacob Hahn, Rice University → Chicago Cubs


Andrew Schaefer, Rice University

Powers S, Ramani S, Hahn J, Schaefer AJ (2026)
“Lead distance under a pickoff limit in Major League Baseball: A sequential game model”
arXiv:2601.15608

Background



- In 2023, MLB limited the pitcher to 3 pickoff attempts
- After 3 failed pickoffs, the runner advances automatically

(video)

The Puzzle

If you are the runner, how much should you extend your lead after each pickoff attempt?

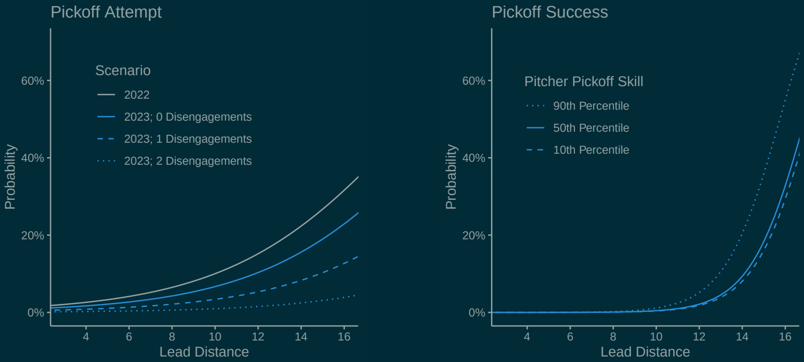
Data: pitch-by-pitch lead distance (player tracking) from MLB*

Our Approach

Step 1: Estimate the effect of lead distance

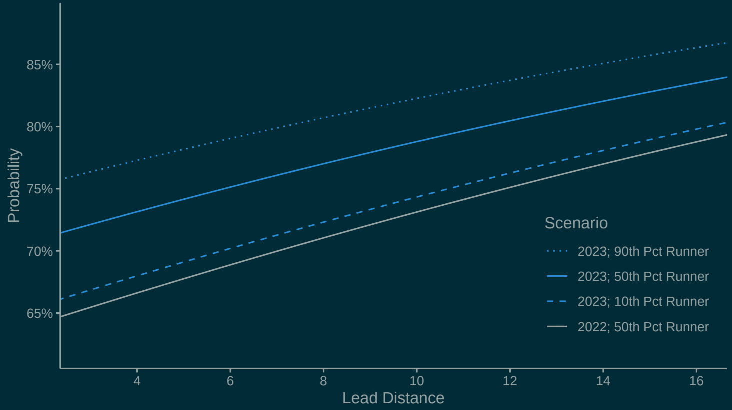
- Linear mixed-effects models (LMMs) for probabilities of pickoff attempt, pickoff success, stolen base success

Pickoff attempts increase with lead distance



- Fixed effects: **lead distance**, **disengagements**, outs, count, runner sprint speed, catcher arm strength
- Random effects: runner, catcher, **pitcher**

Stolen bases increase with lead distance



- Fixed effects: lead distance, disengagements, outs, count, runner sprint speed, catcher arm strength
- Random effects: runner, catcher, pitcher

Solving for Optimal Lead Distance

Two options:

1. Two-player stochastic game
   - Runner chooses lead distance
   - Pitcher chooses whether to attempt pickoff
2. Single-player Markov decision process (MDP)
   - Runner chooses lead distance
   - Pitcher follows known stochastic response (no agency)

The Two-Foot Rule

Count	Disengagements 0	1	2
0-0	10.0	11.5	14.1
1-0	10.2	11.7	14.2
2-0	10.2	11.7	14.2
3-0	10.3	11.8	14.2
0-1	9.8	11.4	14.1
1-1	10.0	11.6	14.2
2-1	10.3	11.8	14.3
3-1	10.4	11.9	14.3
0-2	9.5	11.1	14.1
1-2	9.9	11.4	14.3
2-2	10.0	11.5	14.2
3-2	10.4	11.9	14.4

- Increase lead by **two feet** after each disengagement
- Value of implementation: ∼12 runs / team / season (∼$12,000,000)

Should you just swing as hard as you can?

Ron Yurko, Carnegie Mellon University

Powers S, Yurko R (2026) "Swinging, Fast and Slow: Interpreting variation in baseball swing tracking metrics"
arXiv:2507.01238, to appear in *The American Statistician*

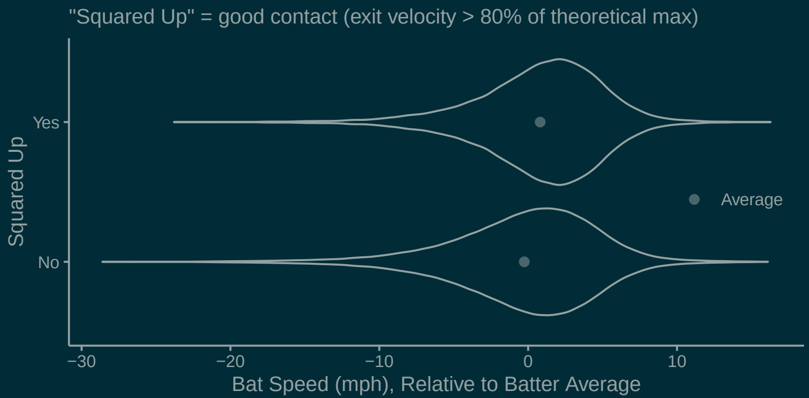
Background

Fans lament batters not having **two-strike approaches**

(video)

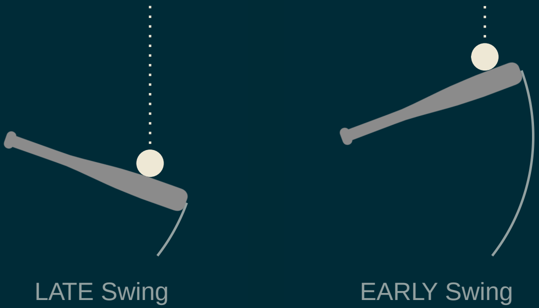
May 2024: MLB releases **swing length** and **bat speed** data

Faster swings make better contact?



Key Insight

We can’t take swing metrics at **face value**



Baseball Savant reports swing metrics at **point of contact**

Step 1: Infer batter intention

What does the batter's swing look like
when they make solid contact?

Bayesian hierarchical model

- skew-normal likelihood for bat speed and swing length
- fixed effects: count, location
- random effects: batter, batter $\times$ count, batter $\times$ location

Some batters shorten/slow their swings more

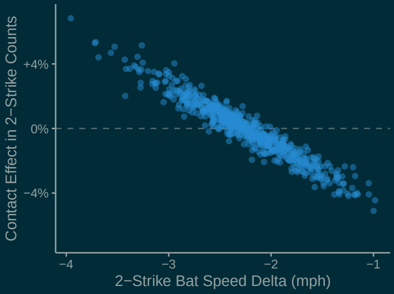
Step 2: Estimate effect of intention

If you make your swing **shorter/slower** with two strikes, do you make **more contact** with two strikes?

Instrumental variables regression (causal inference)

- logistic regression for contact, linear regression for power
- instrument: count
- treatment: batter $\times$ count random effect
- control for difficulty of pitch (gradient boosting model)

Slower swings mean more contact, less power



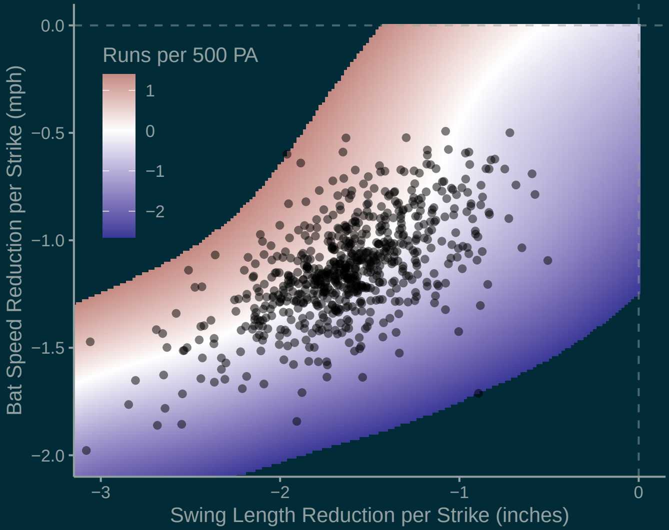
Step 3: Value the contact-power tradeoff

Is **increasing** your contact rate worth **decreasing** your power?

Markov chain model

- state space: ball-strike count
- transition probabilities: depend on approach
- value function: linear weight of plate appearance outcome

The tradeoff is (roughly) balanced

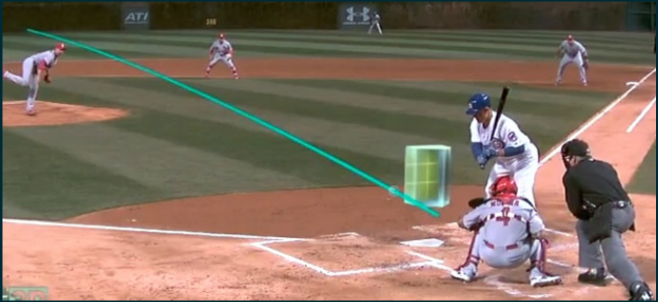


What does a pitch tell you about a pitcher?

 Vicente Iglesias, Susquehanna International Group

Working paper:
Powers S, Iglesias V "Pitch trajectory density estimation for predicting future outcomes"

Background

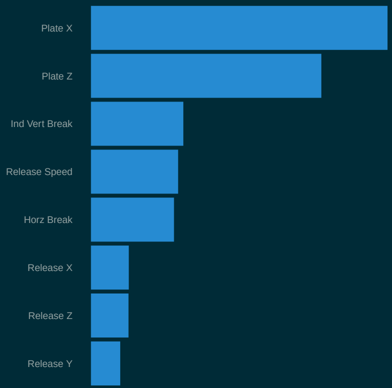


- Full pitch tracking data have been available for 20 years
- Standard practice: Evaluate pitchers based on trajectories

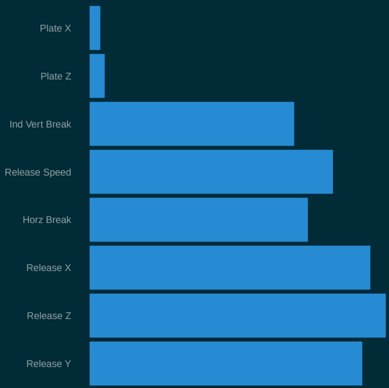
(video #1) (video #2)

Evaluating the *pitch* $\neq$ evaluating the *pitcher*

Variable Importance[1]



Variable Reliability[2]



[1] fractional contribution of each feature's splits to gradient boosting pitch model
[2] (between-pitcher variance) / (total variance); varies by pitch type (here: RHB FB)

Key Insight

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
x_2	$\rightarrow$	y_2
x_3	$\rightarrow$	y_3
	...	
x_n	$\rightarrow$	y_n
x^*	$\rightarrow$	$\hat{y} = \hat{f}(x)$

Our Problem

Pitcher	Pitch		Outcome
A	x_1	$\rightarrow$	y_1
A	x_2	$\rightarrow$	y_2
B	x_3	$\rightarrow$	y_3
		...	
C	x_n	$\rightarrow$	y_n
		$\searrow$	
A			$\hat{y}$

Key Insight

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
x_2	$\rightarrow$	y_2
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	...	
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Our Approach

Pitcher	Pitch		Outcome
A	x_1	$\rightarrow$	y_1
A	x_2	$\rightarrow$	y_2
B	x_3	$\rightarrow$	y_3
		...	
C	x_n	$\rightarrow$	y_n
	$\downarrow$		
A	$\hat{p}(x\|A)$	$\rightarrow$	$\int \hat{p}(x)\hat{f}(x)$

Our Approach

Step 1: Estimate trajectory distributions conditional on covariates

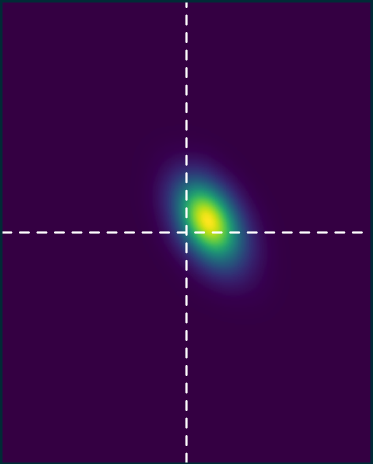
- Bayesian hierarchical model (9D MVN likelihood)

Step 2: Fit supervised learning model and integrate w.r.t. the distribution model

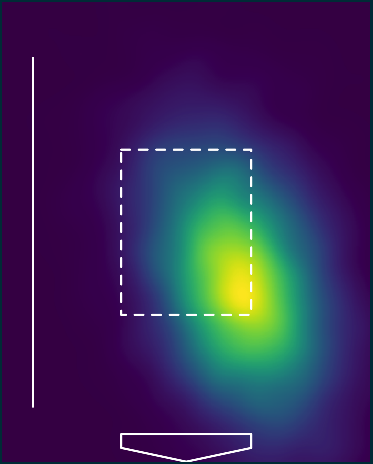
- Gradient boosting

Dylan Cease's Slider vs RHB in 0-0 Counts

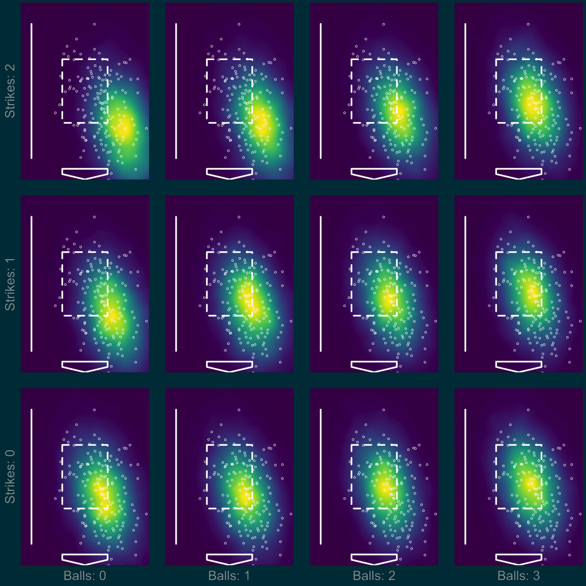
Predicted Break Chart



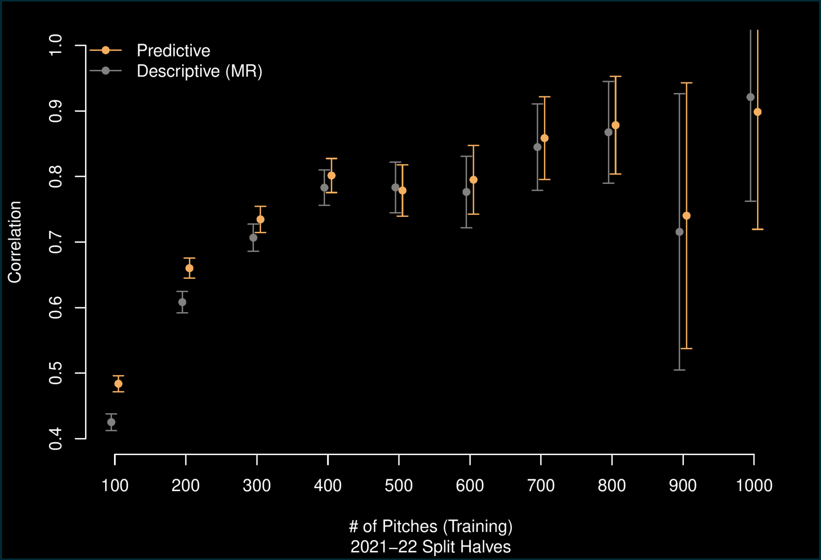
Predicted Plate Location



Dylan Cease's Slider vs RHB in All Counts



Predictive Performance



Thank You!

saberpowers.github.io/research

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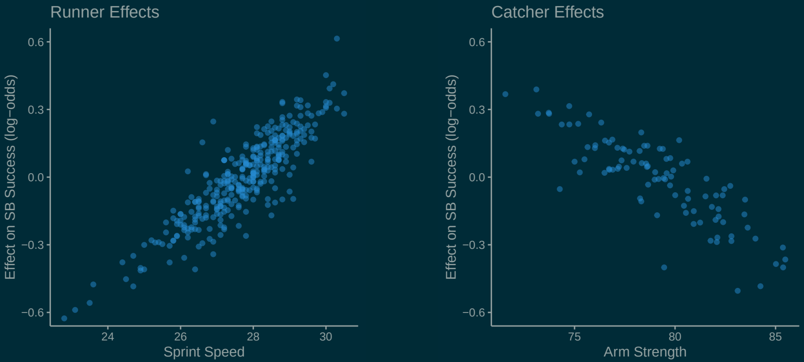
Appendix

Runner and Catcher Effects



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Challenge #2: Confounding